

GRADE 11 BIOLOGY - PROBABLE 8-MARK LONG QUESTIONS

Pioneer Basic Biology (Gurung & Ghimire) - Full syllabus: Cell Biology - Bacteria - Protista - Plasmodium
- Paramecium - Annelida + Bacterial Nutrition

16 probable long questions, each with a complete **8-mark model answer written BELOW the question** + the diagram you must draw + examiner tips. Revise one page a day.

How to SCORE an 8-mark answer (read first)

The 8-mark recipe: 1 labelled diagram (worth ~2 marks) + 5-7 bullet points, each 1-2 lines (1 mark per solid point) + 1 closing line linking the topic to its significance (bonus mark). **Always:** (a) underline the key terms, (b) write the 3 keys/classification if it's a classify question, (c) add one example for each point, (d) draw the diagram on the side, label every part with a straight pointer line, title the diagram at the bottom.

Index of probable questions (by syllabus section):

#	Section	Probable 8-mark question
1-3	Cell Biology	Prokaryote vs eukaryote - Eukaryotic cell structure & organelles - Fluid-mosaic membrane
4-6	Bacteria	Bacterial cell structure + Bact vs Archaea - Nutrition in bacteria (classified) - Reproduction + recombination
7-9	Protista	4 classes of protozoa - Euglena structure + mixotrophy - Beneficial & harmful protozoa
10-11	Plasmodium	Complete life cycle - 5 species + fever pattern + control
12-13	Paramecium	Structure & functions (2 nuclei) - Full life cycle (F-C-E)
14-16	Annelida	Phylum traits + 3 classes - Pheretima external structure - Metamerism + internal systems

Q1 With a well-labelled diagram, compare and contrast the structure of a PROKARYOTIC and a EUKARYOTIC cell. **8 marks**

Earn the 8 marks: 1 drawing of each cell type (2) + 4 differences (4) + 1 similarity line + significance (2).

ANSWER (write this under the question)

- **1. Prokaryote (bacteria):** smallest living cells, 0.2-10 μm ; NO true nucleus - a naked circular DNA called nucleoid (genophore) without nuclear membrane, no histones.
- **2. Eukaryote (plant/animal):** 10-100 μm ; TRUE nucleus - DNA + histones inside a double nuclear envelope with pores + a nucleolus.
- **3. Organelles:** prokaryote has NO membrane-bound organelles (no mitochondria, Golgi, ER, plastids); eukaryote has ALL of these plus a cytoskeleton.
- **4. Ribosomes:** prokaryote = 70S (30S + 50S); eukaryote = 80S in cytoplasm (but 70S inside its mitochondria & chloroplasts).
- **5. Cell wall & respiration:** prokaryote wall = murein/peptidoglycan (absent in Mycoplasma); respiration occurs on MESOSOME folds; eukaryote wall = cellulose (plant) or none (animal); respiration in mitochondria.
- **6. Division:** prokaryote divides by binary fission (a mitosis-less split); eukaryote by mitosis (somatic) / meiosis (germ).
- **7. Similarity (must write!):** both have a plasma membrane, cytoplasm, ribosomes, DNA as genetic material and store glycogen/fat.
- **8. Significance:** prokaryotes are the oldest, simplest life (decomposers/mineralizers of the biosphere); eukaryotes built the complex multicellular world.

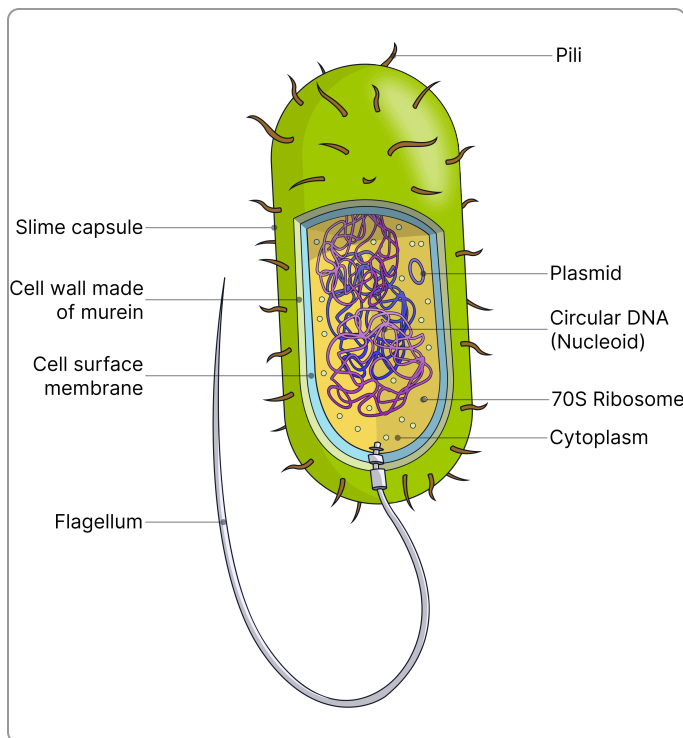


Diagram 1a: Prokaryotic cell - label: nucleoid, murein wall, 70S ribosome, mesosome, flagellum, plasmid, gas vacuole

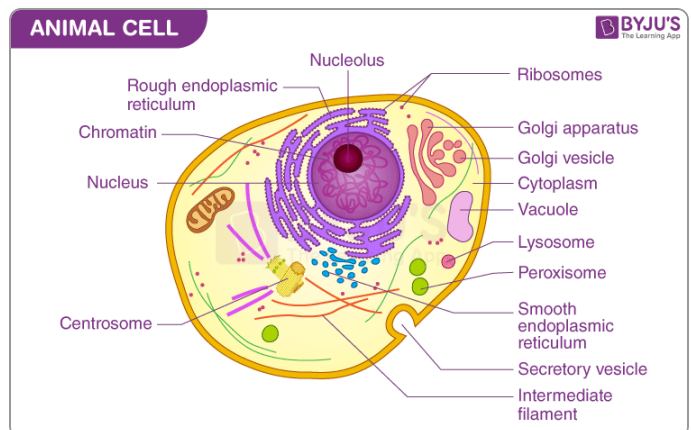


Diagram 1b: Eukaryotic cell - label: nucleus, mitochondria, RER/SER, Golgi, lysosome, centriole, plasma membrane

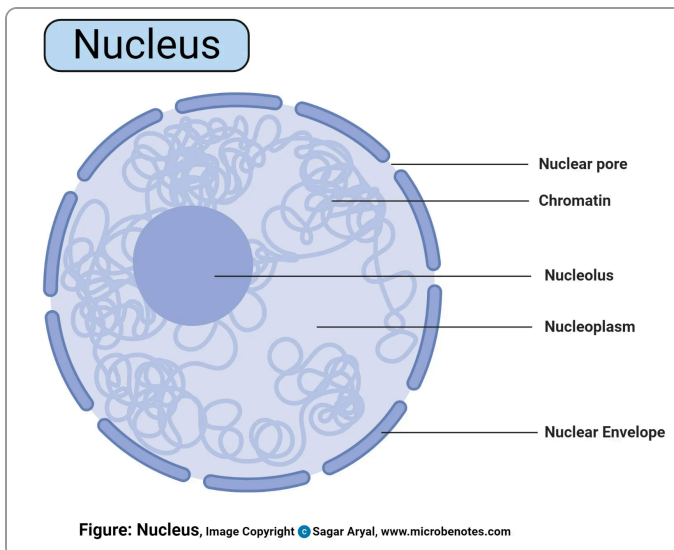
Examiner tip: the single most-tested fact = "70S ribosomes + murein wall + nucleoid" for prokaryote; "80S + true nucleus + mitochondria" for eukaryote. A 4-row difference table also scores full marks.

Q2 Draw a well-labelled diagram of a eukaryotic cell and describe the structure and function of the main organelles. **8 marks**

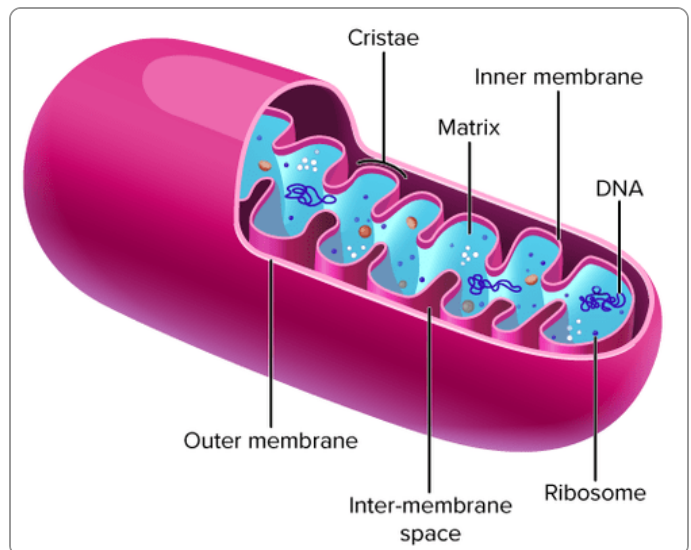
Earn the 8 marks: 1 big labelled cell diagram (3) + 5 organelles, each: structure + function (4) + 1 line on cell size/scales (1).

ANSWER

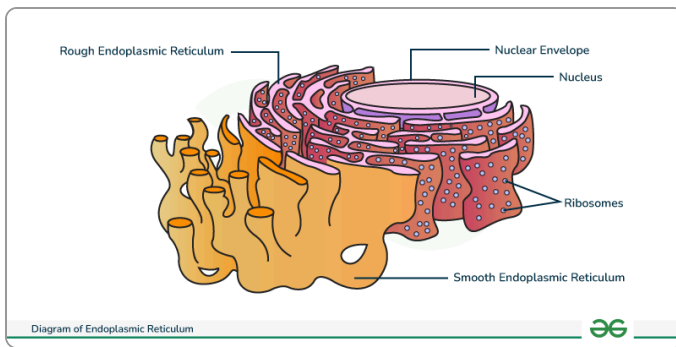
- **1. Nucleus:** double membrane with 300-1000 pores + nucleolus (ribosome factory) + chromatin; control centre of the cell.
- **2. Mitochondria:** double membrane; inner membrane folds = cristae; matrix holds 70S ribosomes + its own DNA; "power house" - ATP by respiration; semi-autonomous.
- **3. ER:** RER (ribosome-studded, makes proteins) and SER (no ribosomes, makes lipids, detoxifies); continuous network of flattened sacs.
- **4. Golgi (dictyosome in plants):** stacked flattened cisternae; modifies, packages and ships proteins; makes the cell wall plates in plants.
- **5. Lysosome:** spherical vesicle of ~40 acid hydrolases (pH 5); digests worn organelles (autophagy) and food (heterophagy); "suicide bag" if it ruptures.
- **6. Ribosomes (80S):** 40S + 60S subunits, free or attached to RER; site of protein synthesis; many form polysomes.
- **7. Others (name them):** peroxisome (catalase breaks H_2O_2 ; plant form = glyoxysome), centriole (9 triplet microtubules; makes spindle; animal only), vacuole + tonoplast (large in plants).
- **8. Scale line:** typical animal cell 20-40 μm ; human body ~ 10 cells; the nucleus (5-10 μm) is the largest organelle.



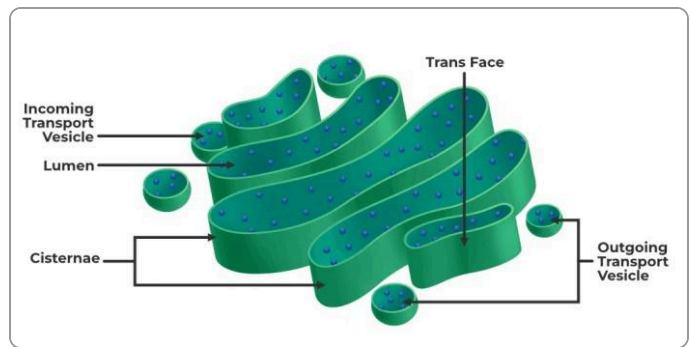
Nucleus - label: nuclear envelope, pores, nucleolus, chromatin



Mitochondrion - label: outer/inner membrane, cristae, matrix, 70S ribosomes



Rough vs smooth ER



Golgi apparatus - label: cis face, trans face, vesicles

Examiner tip: if only space for 2 diagrams: draw (a) the whole cell and (b) one organelle close-up. Always say "semi-autonomous" for mitochondria & chloroplasts - it earns its own mark.

Q3 Explain the fluid-mosaic model of the cell membrane with a well-labelled diagram. **8 marks**

Earn the 8 marks: 1 membrane diagram (2) + 4 structural components (4) + 2 functional lines (2).

ANSWER

- **1. What it is:** Singer & Nicolson (1972) model - the membrane is a FLEXIBLE LIPID BILAYER in which PROTEINS are embedded like a mosaic, free to drift ("fluid").
- **2. Phospholipid bilayer:** two layers; each phospholipid has a hydrophilic (water-loving) head out + hydrophobic (water-fearing) tails in.
- **3. Proteins:** INTEGRAL proteins (span the bilayer; act as channels & pumps) + PERIPHERAL proteins (sit on the surface; attachment/signalling).
- **4. Cholesterol:** squeezed between the tails - stabilizes the membrane and reduces fluidity at high temperature.
- **5. Glycoproteins/glycolipids:** sugar + protein/lipid on the outer face - cell recognition, antigen sites, receptor sites.
- **6. Function 1 - selective permeability::** small non-polar molecules diffuse freely; ions & large molecules need protein channels/pumps (passive vs active transport).
- **7. Function 2 - transport + recognition + adhesion:** carrier proteins move nutrients; glycoproteins identify "self vs foreign"; membrane fuses for endo/exocytosis.
- **8. Boundary & limit:** membrane thickness ~ 7-8 nm; the cell coagulates (denatures) at 60degC; it is the "living" membrane - a dead membrane is merely semipermeable.

Fluid Mosaic Model

Cell Membrane

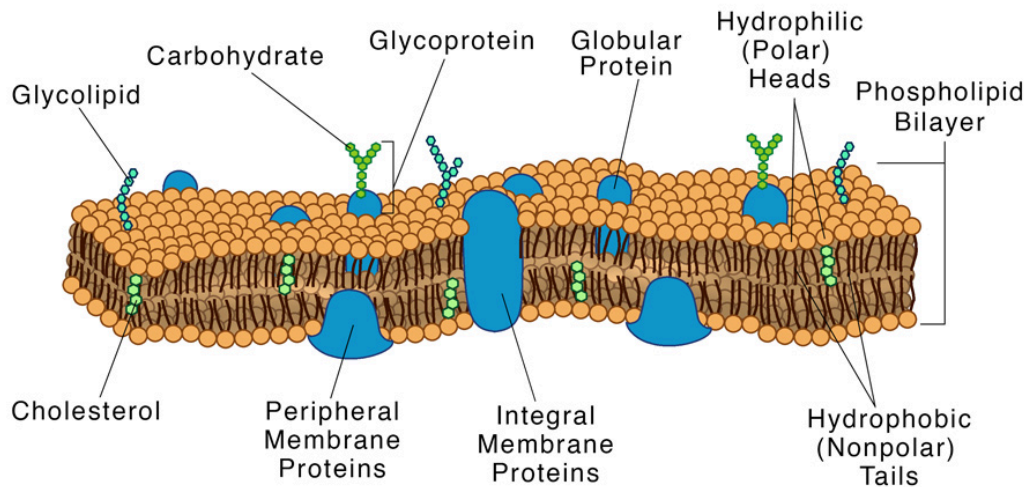


Diagram: fluid-mosaic model - label: phospholipid bilayer, integral protein (channel), peripheral protein, cholesterol, glycoprotein (sugar head out)

Examiner tip: "Fluid" = lipids + proteins can move laterally; "Mosaic" = many different proteins scattered in. Write both words in the first line - it is the question.

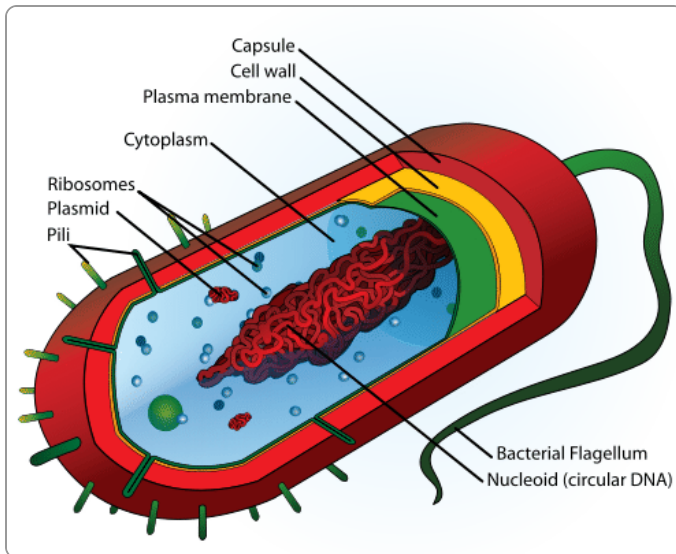
Q4 Describe the structure of a bacterial cell with a well-labelled diagram and distinguish bacteria from archaea. **8 marks**

Earn the 8 marks: 1 bacterial-cell diagram (3) + 5 prokaryote structure points (4) + Bacteria-vs-Archaea table (1).

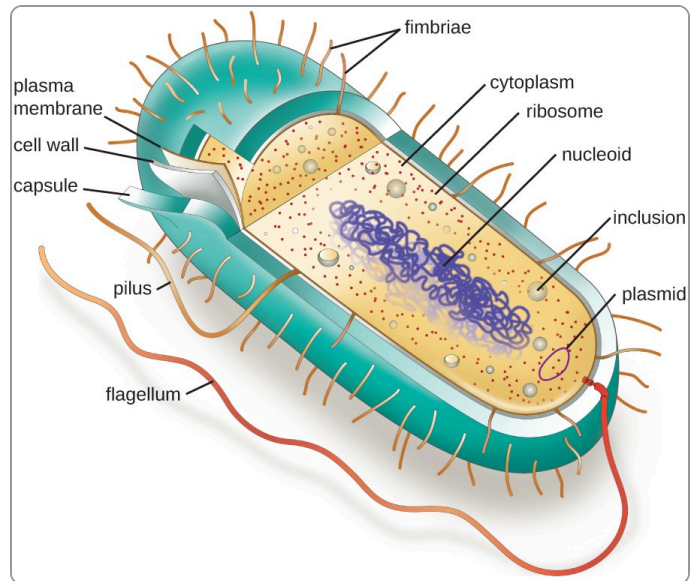
ANSWER

- **1. Identity:** bacteria are the simplest, smallest MICROSCOPIC PROKARYOTIC cells (Kingdom Monera); no true nucleus or membrane-bound organelles.
- **2. Nucleoid (genophore):** circular DNA + RNA + protein, NO histone protein, no nuclear membrane - the "incipient nucleus".
- **3. Cell wall:** rigid MUREIN (mucopeptide / peptidoglycan) of N-acetylglucosamine + N-acetylmuramic acid - ABSENT in Mycoplasma.
- **4. Ribosomes 70S (30S + 50S);** gas vacuoles for buoyancy in aquatic types; no true vacuole; reserve food = glycogen / fat; flagella of protein flagellin; MESOSOME = respiration fold.
- **5. Division & recombination:** binary fission; no true sex - recombination by conjugation / transformation / transduction.
- **6. Nutrition:** autotrophic or heterotrophic; they are the decomposers & mineralizers of the biosphere.

- **7. Bacteria vs Archaea (table):** wall = murein vs pseudomurein; membrane lipids = ESTER vs ETHER bonds; bacteria grow everywhere, archaea favour EXTREME habitats (hot springs, salt lakes) + methane producers.
- **8. Examples:** Vibrio, Diplococcus, Bacillus, Streptococcus, E. coli; Mycoplasma = smallest living cell (0.1-0.3 μm).



Bacterial structure + shapes (coccus, bacillus, vibrio, spiral)



Unique prokaryote features: murein, 70S, plasmids

Examiner tip: "murein / 70S / circular DNA no histone" are the 3 exam markers of a bacterium. Don't call Mycoplasma the "smallest living thing" - viruses are smaller but not living.

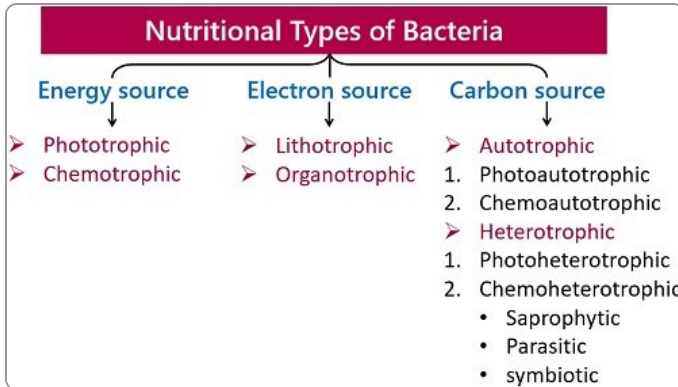
Q5 Classify bacteria according to their NUTRITION and give one example of each type. **8 marks**

Earn the 8 marks: 1 classification diagram (3) + the 4 types, each with source + example (4) + 1 O₂-requirement line (1).

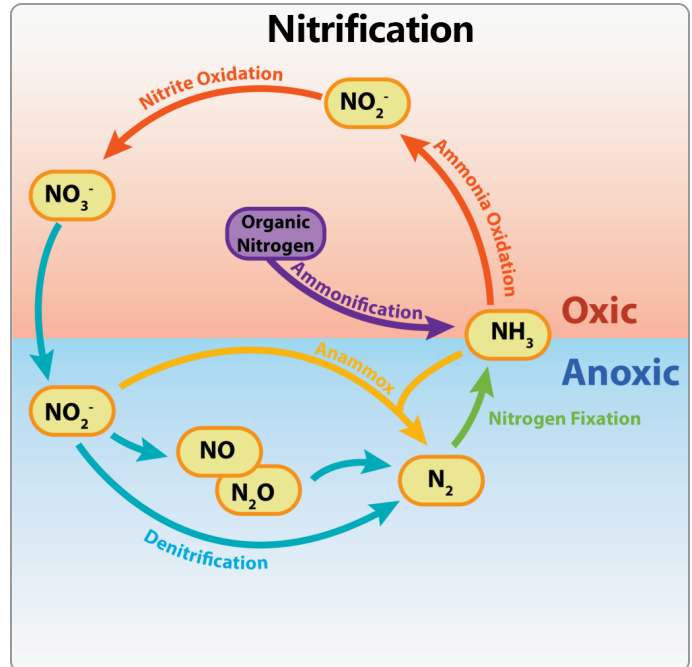
ANSWER

- **1. The 3 keys ("E-C-E"):** ENERGY (light / chemical) - CARBON (CO₂ / organic) - ELECTRON DONOR (inorganic / organic). A bacterium's type = its 3 answers.
- **2. Photoautotrophs** (light + CO₂): Cyanobacteria (*oxygenic, releases O₂*); green & purple sulfur bacteria (*anoxygenic, use H₂S*).
- **3. Chemoautotrophs / chemolithotrophs** (inorganic energy + CO₂): nitrifying *Nitrosomonas, Nitrobacter*; sulfur *Thiobacillus, Beggiatoa*; iron *Ferrobacillus*.
- **4. Chemoorganotrophs / chemoheterotrophs** (organic energy + C): the most common - *E. coli, Staphylococcus, Lactobacillus*; lifestyles: saprophyte / parasite / symbiont.
- **5. Photoheterotrophs** (light + organic C): purple non-sulfur (*Rhodospirillum*) & green non-sulfur (*Chloroflexus*).

- **6. O₂ axis:** each is also an obligate aerobe / obligate anaerobe / facultative anaerobe / microaerophile / aerotolerant (e.g. *E. coli* = facultative, *Clostridium* = obligate anaerobe).
- **7. Metabolic fates of glucose:** aerobic respiration (O₂, ~30-38 ATP) / anaerobic respiration (NO₃-acceptor) / fermentation (2 ATP, lactic/ethanol).
- **8. Significance:** these bacteria fix carbon, nitrify (N-cycle), denitrify, fix N₂ and decompose - the biosphere's chemical engine.



Bacterial nutrition - the 4 nutritional types



Chemolithotrophy & N-metabolism (nitrifiers, S-bacteria)

Examiner tip: memorize "PHL / CTL / CHO / PHL": PHL = photo-litho-AUTO (cyanobacteria), CTL = chemo-litho-AUTO (nitrifiers), CHO = chemo-organo-HETERO (most bugs). "Obligate = forced; facultative = can switch."

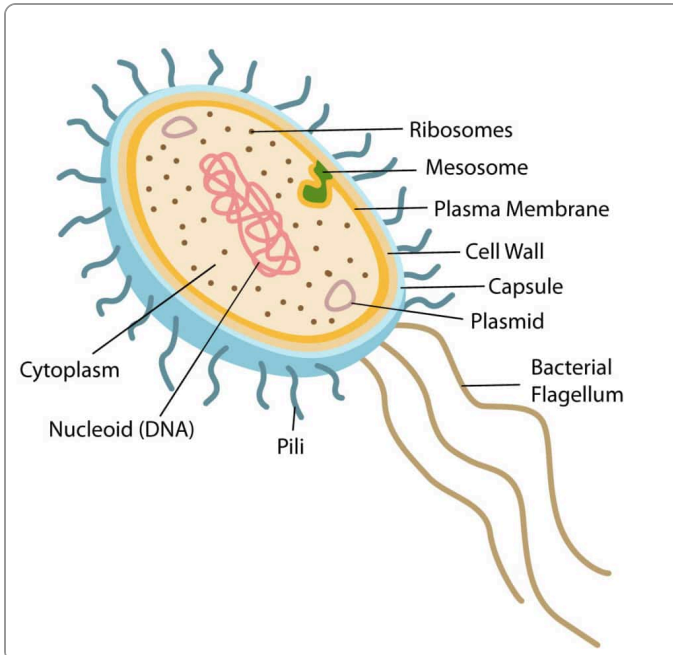
Q6 Explain the modes of reproduction in bacteria, including asexual division and the 3 types of genetic recombination. **8 marks**

Earn the 8 marks: 1 fission/shape diagram (2) + 3 recombination types, each described (4) + 1 "no true sex" + significance line (2).

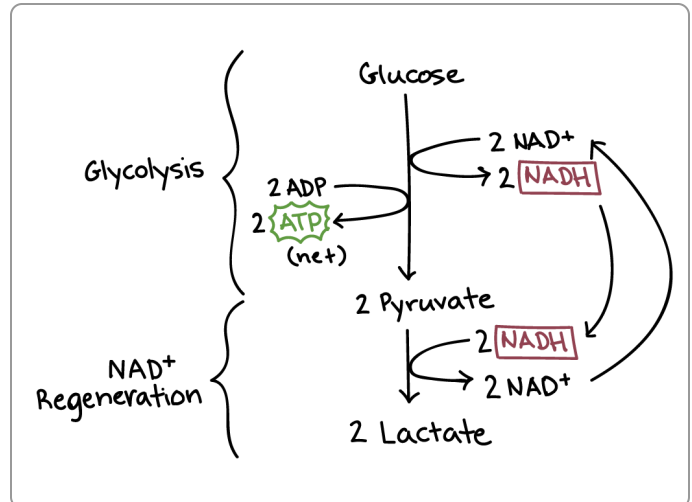
ANSWER

- **1. Binary fission (asexual, main):** circular DNA replicates, cell elongates, septum forms -> two IDENTICAL daughter cells (clones); *E. coli* division time ~ 20 min.
- **2. Endospore formation:** a SURVIVAL structure (NOT reproduction) of *Bacillus* & *Clostridium*; heat/acid-resistant, germinates when water returns (1 cell -> 1 spore -> 1 cell).
- **3. CONJUGATION:** an F⁺ cell transfers an F-plasmid (fertility) to an F⁻ cell through a SEX PILUS; the recipient becomes F⁺ (plasmid transfer, not whole-genome).

- **4. TRANSFORMATION:** a cell takes up **FREE (naked) DNA** from the environment (Griffith 1928) and incorporates it - basis of gene engineering.
- **5. TRANSDUCTION:** a **BACTERIOPHAGE (virus)** accidentally carries a donor's gene into a new host cell during infection.
- **6. Mnemonic "CTC":** Conjugation (pilus) - Transformation (naked DNA) - Transduction (phage transit) - the **ONLY** genetic mixing in prokaryotes.
- **7. No true sex:** bacteria show no syngamy / meiosis; recombination just shuffles genes without making more cells.
- **8. Significance:** recombination gives variability + antibiotic-resistance transfer (via plasmids & transposons "jumping genes").



Prokaryotic cell - draw & label for the fission stage



Bacterial metabolic / division pathways context

Examiner tip: always write "CTC" and one-line each. Trap: "endospore = reproduction" is **WRONG** - it's survival. "Binary fission = 1 -> 2 clones".

Q7 Describe the 4 classes of protozoans (Kingdom Protista) on the basis of locomotion, with examples. **8 marks**

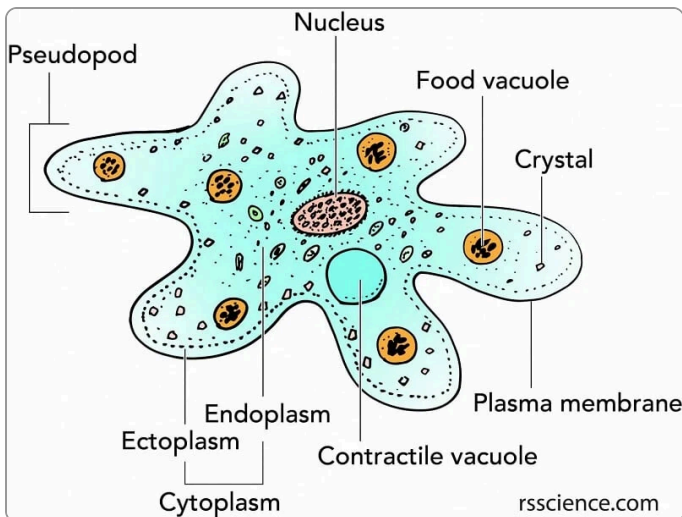
Earn the 8 marks: 1 protozoan diagram (2) + 4 classes, each: locomotion + example + one feature (4) + 1 algal-division + lifestyle line (2).

ANSWER

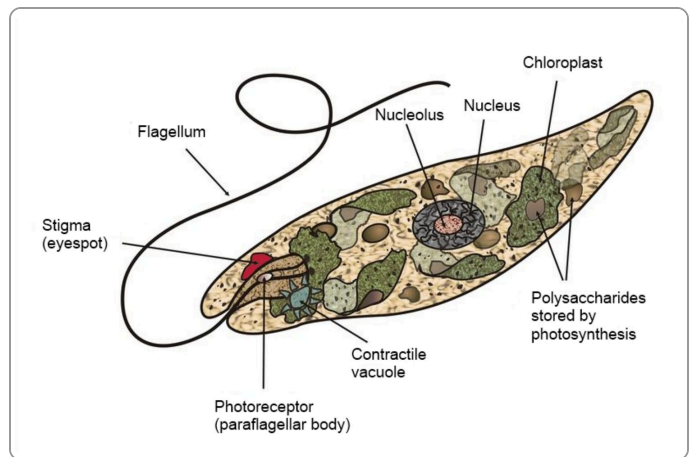
- **1. Kingdom Protista:** eukaryotic "residual" unicells - not plant/animal/fungus. Coined by Haeckel (1869); mostly microscopic, free-living or parasitic.
- **2. SARCODINA (Rhizopoda / Amoeboid):** move by pseudopodia (false feet, cytoplasmic streaming); irregular shape; phagocytic; e.g. *Amoeba proteus*, *Entamoeba histolytica* (causes

dysentery).

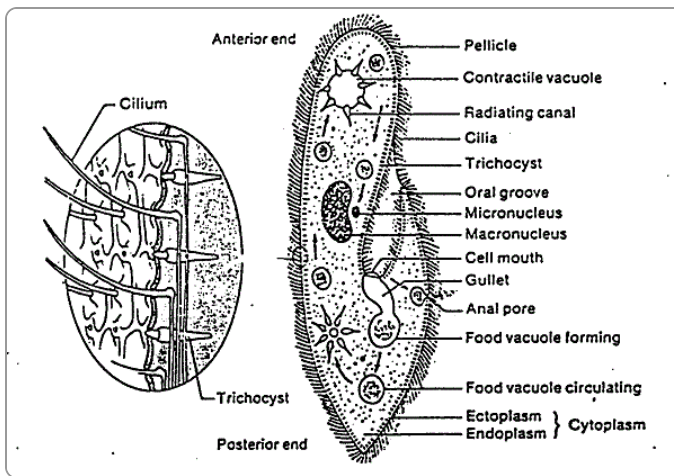
- **3. MASTIGOPHORA (Flagellate):** move by flagella (1 to many); a flexible plicella; some have chloroplasts; e.g. *Euglena* (mixotroph, red eyespot), *Trypanosoma* (sleeping sickness), *Giardia*.
- **4. CILIOPHORA (Infusoria / Ciliate):** move by cilia (short, many, rhythmic); the **MOST COMPLEX** protozoa; **TWO** nuclei (macro + micro); e.g. *Paramecium caudatum*, *Stentor* (see Q12-13).
- **5. SPOROZOA (Apicomplexa):** move by none in the adult - obligate **PARASITES**, spore-forming; apical complex for invasion; e.g. *Plasmodium* (malaria), *Toxoplasma* (see Q10-11).
- **6. Mnemonic "P-F-C-N":** Pseudopod / Flagellum / Cilia / None - one locomotion type per class = the identifying key.
- **7. Nutrition types:** holozoic (ingest - ciliates), osmotrophic (absorb - slime mould), photosynthetic / mixotrophic (*Euglena*, algae).
- **8. The 2 algal divisions:** many books list *Euglenophyta* / *Chrysophyta* (golden-brown, diatom siliceous test) as the "plant" halves - **CHECK** your book's exact 6 names.



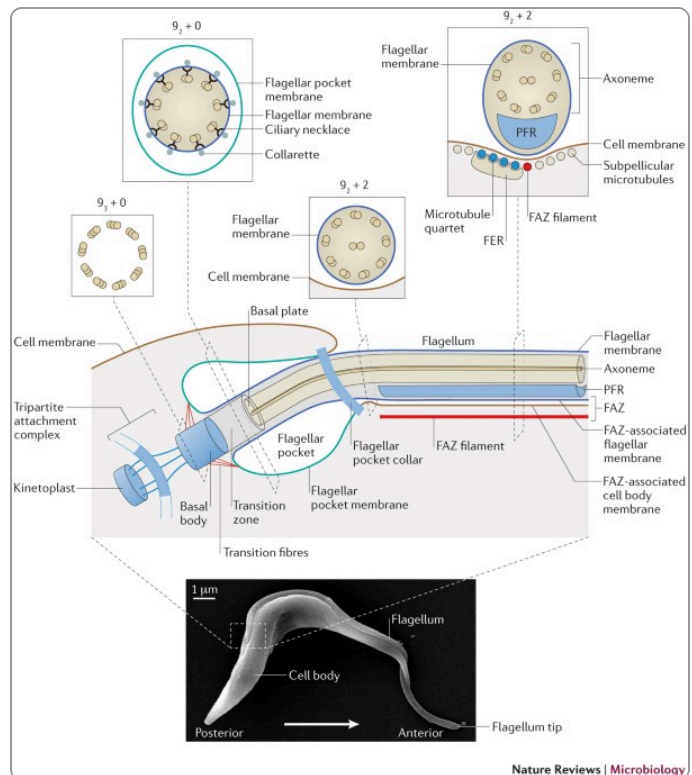
Sarcodina - Amoeba (pseudopodia, food vacuole, nucleus)



Mastigophora - Euglena (flagellum, eyespot, chloroplast)



Ciliophora - Paramecium (cilia, oral groove, 2 nuclei)



Mastigophora - Trypanosoma (flagellum + undulating membrane)

Examiner tip: the class = the locomotion organ. Write "P-F-C-N" first, then one example + one extra feature each. Ciliophora's TWO nuclei is its signature.

Q8 Describe the structure, nutrition and reproduction of *Euglena* with a well-labelled diagram. **8 marks**

Earn the 8 marks: 1 *Euglena* diagram (3) + 5 structure parts (3) + nutrition + reproduction lines (2).

ANSWER

- **1. Identity:** *Euglena gracilis* = a MASTIGOPHORAN protozoan; slipper-shaped, 60-70 µm; lives in stagnant, plant-rich water; both a "protozoan" and a "photosynthetic alga" (bridges both worlds).
- **2. Pellicle:** a thin flexible covering (not a rigid wall) under the membrane - gives the body shape + allows contraction (metaboly).
- **3. Flagellum:** one to several long whip-like flagella = locomotion (swimming).
- **4. Chloroplast + eyespot:** chloroplasts do photosynthesis; the red "eyespot" (stigma) senses light direction so the cell swims toward it.
- **5. Nucleus + contractile vacuole:** one true nucleus; a pulsatile (contractile) vacuole expels excess water (osmoregulation).
- **6. Nutrition = MIXOTROPHY:** in light it is PHOTOSYNTHETIC (plant-like); in dark it is HETEROTROPHIC (absorbs dissolved organics) - "mixotrophic" is the key word.

- **7. Reproduction:** asexual by longitudinal (or transverse) binary fission; some forms a resistant cyst (epilimnium / hypnospore).
- **8. Significance:** a model for studying plant-animal boundaries; used in wastewater treatment & as a protein feed; shows the "residual" nature of Protista.

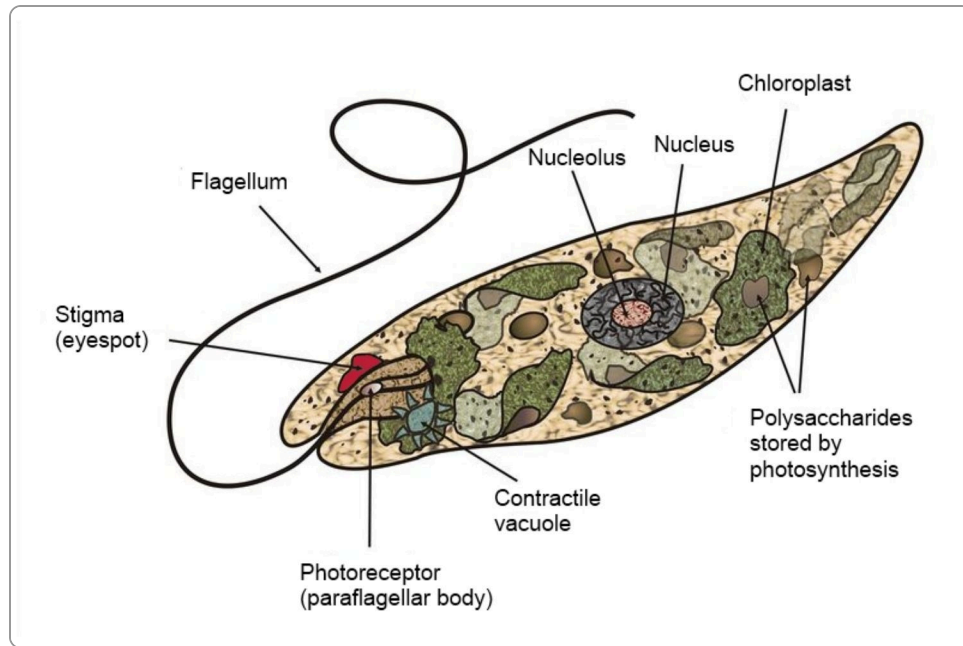


Diagram: *Euglena* - label: pellicle, flagellum, eyespot (stigma), chloroplast, nucleus, contractile vacuole

Examiner tip: "mixotrophic" is the one word that scores the whole nutrition mark - say photosynthetic in light, heterotrophic in dark. Eyespot = red light sensor, NOT an eye.

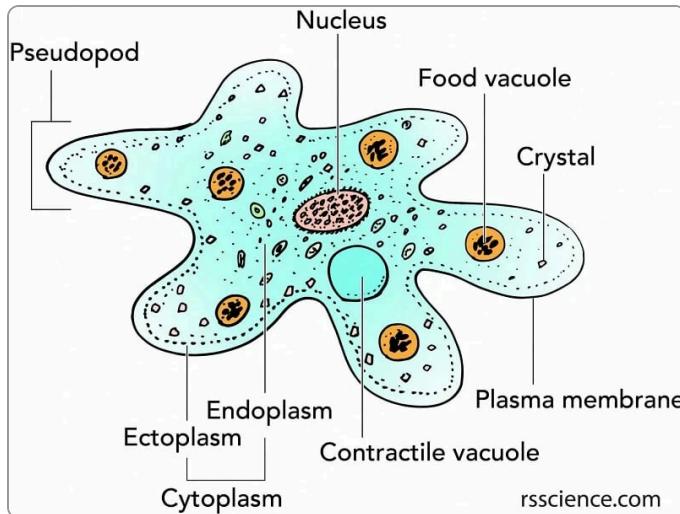
Q9 Write about the beneficial and harmful economic / medical importance of protozoa. **8 marks**

Earn the 8 marks: 2 protozoan diagrams (2) + 4 beneficial uses (3) + 4 harmful (diseases) (3).

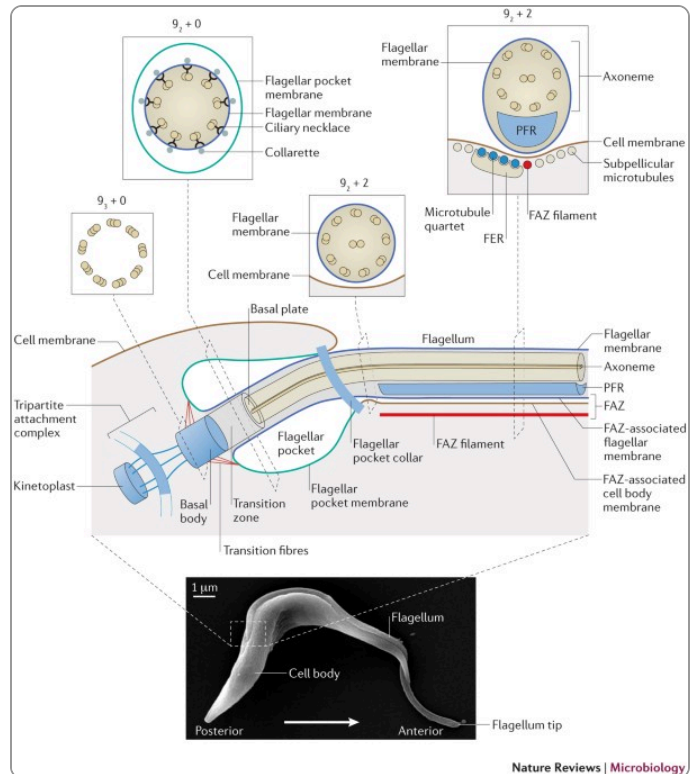
ANSWER

- **1. Kingdom recap:** protozoa = unicellular eukaryotes; many live as free decomposers, some as parasites - so they help AND harm.
- **2. Beneficial 1 - O₂ & food:** photosynthetic protozoa (*Euglena*, diatoms, dinoflagellates) are major O₂ producers & the base of aquatic food chains.
- **3. Beneficial 2 - food & lab work:** paramecia & infusoria feed larvae of fishes, frogs & insects; *Paramecium* is a classic lab study organism.
- **4. Beneficial 3 - water purification & industry:** ciliates in sewage treatment remove bacteria & organics; some are bio-indicators of water quality.
- **5. Harmful 1 - dysentery:** *Entamoeba histolytica* causes amoebic dysentery / liver abscess (fecal-oral).
- **6. Harmful 2 - sleeping sickness & Chagas:** *Trypanosoma brucei* & *T. cruzi* (tsetse / triatomine vectors).

- **7. Harmful 3 - diarrhoea & trichomoniasis:** *Giardia lamblia* (water-borne); *Trichomonas vaginalis* (venereal).
- **8. Harmful 4 - the biggest:** *Plasmodium* causes **MALARIA**, the deadliest human protozoan disease (see Q10-11).



Amoeba / Sarcodina (Entamoeba causes dysentery)



Trypanosoma (sleeping sickness / Chagas)

Examiner tip: pair EACH disease with its protozoan + route. "Plasmodium = malaria, Trypanosoma = sleeping sickness, Entamoeba = dysentery, Giardia = diarrhoea" is the 4-disease mark-list.

Q10 Describe the complete life cycle of *Plasmodium* (malaria) with a well-labelled diagram showing the human and mosquito phases. **8 marks**

Earn the 8 marks: 1 7-step life-cycle diagram (4) + 4 human-phase steps (2) + 4 mosquito-phase + "digenetic" line (2).

ANSWER

Life cycle (write step-by-step, then draw the 7-box cycle):

- **1. Host alternation (digenetic):** asexual phase in the HUMAN + sexual phase in the female *Anopheles* mosquito - the single most important exam fact.
- **2. Infection:** a biting *Anopheles* injects crescentic SPOROZITES into the blood.
- **3. Hepatic (pre-erythrocytic) phase:** sporozoites enter the LIVER, multiply (schizogony) -> merozoites; this is latent/asymptomatic.
- **4. Erythrocytic (blood) phase:** merozoites invade RBCs -> trophozoite -> segmented schizont -> RBC ruptures -> CHILLS + FEVER + anaemia (the malaria attack).

- **5. Gametocytogenesis:** some trophozoites become GAMETOCYTES (round macrogamete / crescent microgamete) - the ONLY form the mosquito can ingest.
- **6. Mosquito phase (sexual):** in the gut the microgamete exflagellates & fertilizes -> ZYGOTE -> motile OOKINETE -> OOCYST (sporogony) -> SPOROZOITES migrate to the salivary glands.
- **7. Reinfection:** the now-infective mosquito bites a new human -> new sporozoites; cycle repeats.
- **8. Latency / control:** *P. vivax* / *ovale* keep HYPNOZOITES dormant in the liver -> RELAPSE; control = drain stagnant water + bed-nets + antimalarials.

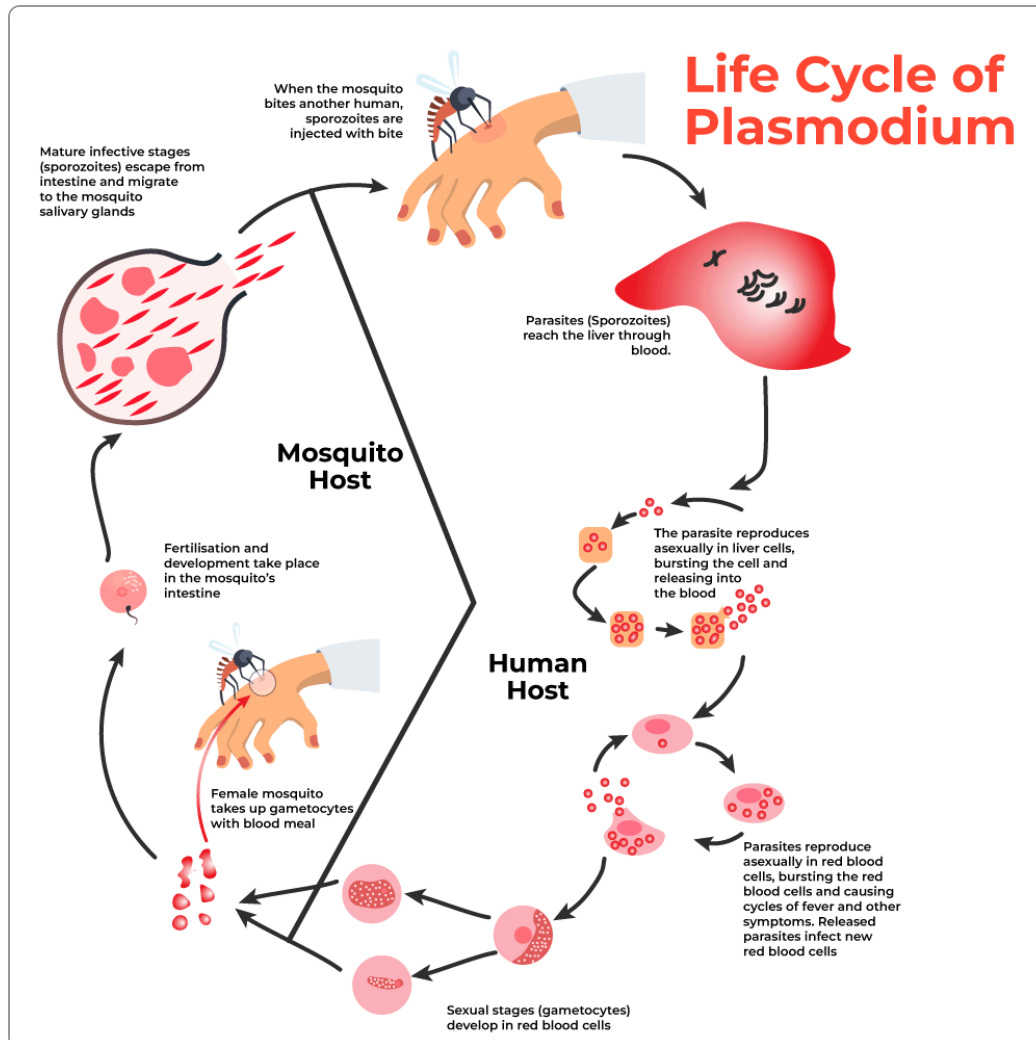


Diagram: 7-box life cycle - label: (human) sporozoite, liver schizont, RBC schizont, gametocyte; (mosquito) zygote, ookinete, oocyst, salivary-gland sporozoites

Mnemonic: "BITE -> LIVER -> BLOOD -> GAMETO -> ZYGOTE -> SALIVA -> BITE" (human steps L+B, mosquito steps Z+S). 4 forms = "S-T-S-G": Sporozoite - Trophozoite - Schizont - Gametocyte.

Examiner tip: colour-code the diagram HUMAN (blue box) vs MOSQUITO (red box). "Sporozoite infects the human; gametocyte infects the mosquito" is the classic mark. falciparum = banana/crescent microgametocyte.

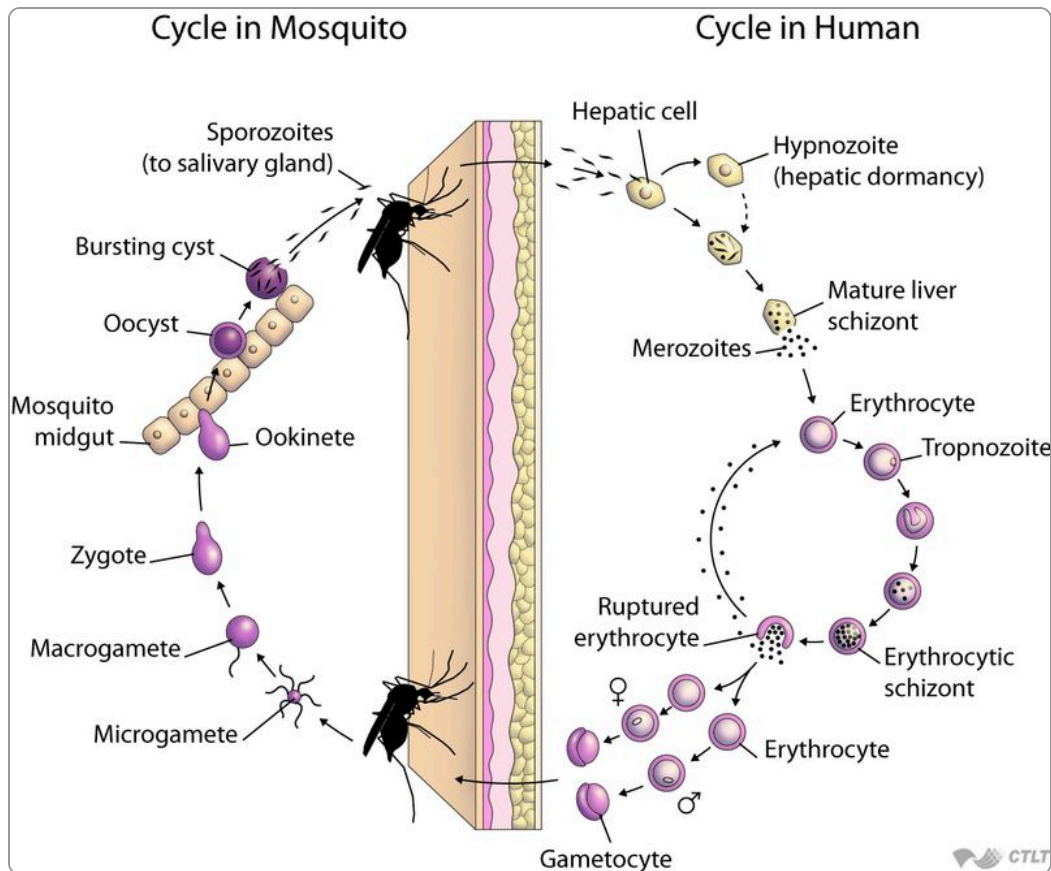
Q11 Differentiate the 5 species of *Plasmodium* by their RBC features and fever pattern, and state control measures. **8 marks**

Earn the 8 marks: 1 species-table (5) + fever-pattern line (1) + 4 control measures (2) + RBC-signature line (1).

ANSWER

Species	Fever period	RBC invaded	Signature mark	Hypnozoite?
<i>P. falciparum</i>	Irregular / malignant	All ages	Crescent (banana) microgametocyte	No
<i>P. vivax</i>	Tertian 48 h (benign)	Oxygenated RBC	Schffner's dots	YES (relapse)
<i>P. ovale</i>	Tertian 48 h	YOUNG RBC	Pointed RBC + dots	YES (relapse)
<i>P. malariae</i>	Quartan 72 h	Old RBC	Rosette / daisy schizont	No
<i>P. knowlesi</i>	Quotidian 24 h	-	Zoonotic (macaque -> man)	-

- **1. Fever naming ("QTTK"):** Q = quartan (*malariae*, 72 h) - TT = tertian (*vivax* & *ovale*, 48 h) - K = 24 h (*knowlesi*); *falciparum* = irregular.
- **2. Relapse:** ONLY *vivax* & *ovale* relapse (hypnozoite in the liver wakes later) - this is their exam identity.
- **3. RBC signature:** *falciparum* = crescent gametocyte; *vivax* = Schffner's dots; *ovale* = pointed RBC; *malariae* = rosette. Diagnosis = thick/thin blood film under the microscope.
- **4. Control (4 measures):** (a) drain stagnant water / prevent breeding; (b) DDT / fogging + bed-nets; (c) antimalarial drugs; (d) vector = female *Anopheles* ONLY, so mosquito control breaks the cycle.
- **5. Significance:** *falciparum* = the most dangerous (cerebral malaria); the 48/72 h periodicity comes from the RBC burst cycles.



Alternate labelled malaria life cycle - use to cross-check the stage names

Examiner tip: "tertian = 48 h, quartan = 72 h, quotidian = 24 h" and "vivax & ovale relapse (hypnozoite)" are the two mark-generators. Draw a 5-row table = full 8.

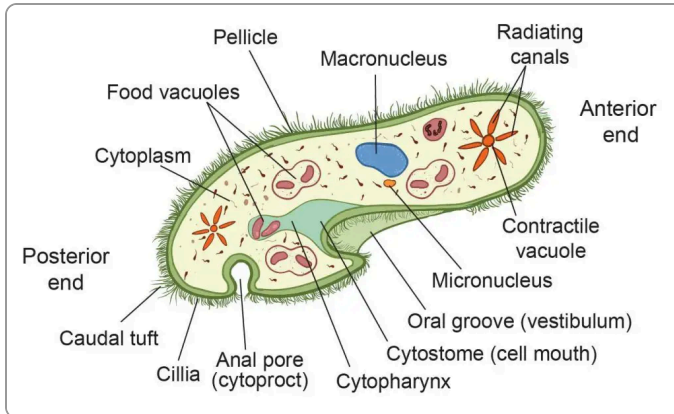
Q12 Draw a well-labelled diagram of *Paramecium caudatum* and describe the structure & function of its main parts. **8 marks**

Earn the 8 marks: 1 big labelled Paramecium diagram (3) + 5 structure parts each with function (4) + 1 identification line (1).

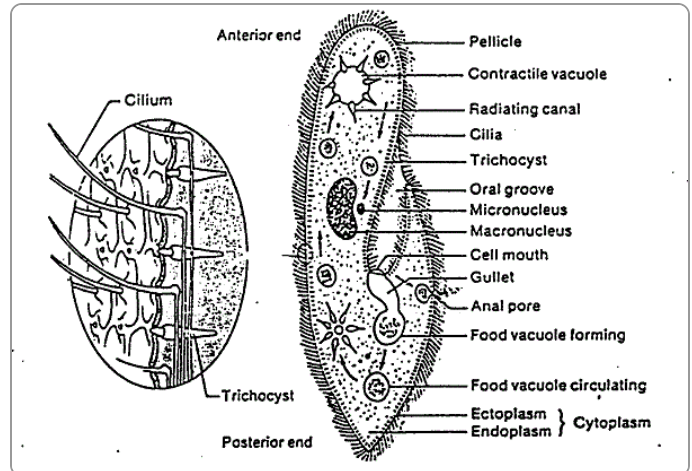
ANSWER

- **1. Identity:** *Paramecium caudatum* = a CILIOPHORAN (Infusoria) free-living freshwater protozoan; slipper-shaped, 50-300 µm.
- **2. Pellicle:** flexible outer covering (no rigid wall) - keeps shape; respiration & excretion happen by diffusion through it.
- **3. Cilia:** thousands of short cilia cover the body - rhythmically beat for LOCOMOTION and to sweep food to the oral groove.
- **4. Oral groove + cytopharynx ("mouth"):** the food funnel; food passes into a FOOD VACUOLE for intracellular digestion; waste exits the CYTOPHARYNX (anal pore) - nutrition = holozoic.
- **5. TWO nuclei (signature!):** the MACRONUCLEUS (somatic - controls food/growth) + the MICRONUCLEUS (germ - used in conjugation, divides by meiosis). "MacroM = Management, MicroM = Meiosis".

- **6. TWO contractile vacuoles:** at the anterior & posterior ends - pump out excess water (OSMOREGULATION, essential in fresh water).
- **7. Respiration & excretion:** no organs - O₂ enters & CO₂ leaves by diffusion across the pellicle; the CV expels liquid waste.
- **8. Identification line:** "many cilia + 2 nuclei + oral groove + 2 contractile vacuoles" is the Ciliophora exam trio - any one missing means not a ciliate.



Paramecium cell structure - label: cilia, oral groove, macronucleus, micronucleus, 2 contractile vacuoles, food vacuole, cytopyg



External + internal features - the 6 parts to label

Examiner tip: the TWO NUCLEI are the one point examiners check hardest. Macro = body/food; Micro = sex/conjugation. "2 CVs + 2 nuclei + oral groove" = the 6-mark label set.

Q13 Describe the complete life cycle of *Paramecium*, including binary fission, conjugation and encystation. **8 marks**

Earn the 8 marks: 1 life-cycle diagram (3) + 3 processes, each with its role (4) + 1 "encystation != reproduction" line (1).

ANSWER

- **1. Asexual - BINARY FISSION (main, "F"):** the 2 nuclei divide (macro by amitosis, micro by meiosis + reformation), the cell splits longitudinally -> two identical daughters = rapid population growth in a healthy pond.
- **2. Sexual - CONJUGATION ("C"):** two cells meet & align; each micronucleus meiotically splits, one haploid is exchanged (reciprocal), a new diploid MACRONUCLEUS forms; the pair separates -> two exconjugants = two NEW VARIETIES. Note: it is RECOMBINATION, NOT "making more cells" (2 in = 2 out).
- **3. Encystation ("E"):** in a dry / starving pond, cilia retract & a thick coat forms -> a resting HYPNOSPORE / METABOLA; when water returns it DORMANT breaks (germinates) -> active ciliate. This is SURVIVAL, not reproduction.
- **4. Mnemonic "F-C-E":** Fission (multiplies) - Conjugation (mixes genes) - Encystation (rests).

- **5. Why conjugation matters:** it restores genetic variation in a clone population; the two new varieties are genetically different from the parents.
- **6. Habitat / trigger:** fission in good food; conjugation when the cells are old / "hungry" (aging triggers it); encystation when the water dries.
- **7. Symbiosis note:** *Paramecium bursaria* hosts green algae (Zoochlorella) inside = protosymbiosis (the cell moves, the algae photosynthesize).
- **8. Significance:** Paramecium is the textbook model for protist reproduction - it shows asexual + sexual + resting all in one organism.

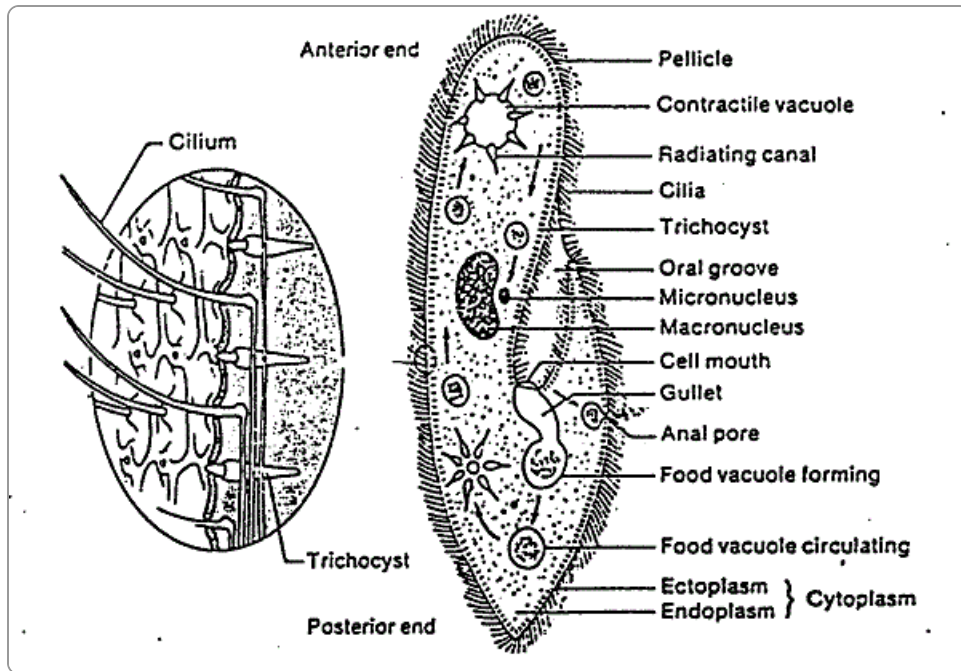


Diagram: 4-box life cycle - Fission (1->2) + Conjugation (2 cells exchange haploid) + Encystation (resting cyst) + germination

Mnemonic: "F-C-E = Fission (more cells) - Conjugation (mix, not more) - Encystation (rest, not more)".
Conjugation = 2 cells in = 2 NEW-VARIETY cells out.

Examiner tip: write "F-C-E" first, then one line each. The mark-gener: "encystation is a resting stage, NOT reproduction" and "conjugation produces new varieties, not daughter cells".

Q14 Describe the phylum Annelida with its characteristics and differentiate its three classes. **8 marks**

Earn the 8 marks: 1 earthworm diagram (3) + 6 phylum characteristics (3) + 3 classes with examples (2).

ANSWER

- **1. Phylum Annelida:** the SEGMENTED worms; ~16,500 species; "annulus" = small ring (each body ring / segment).
- **2. Metamerism:** the body is divided into repeated, similar SEGMENTS (metameres) - each carrying the same set of organs (homomeric metamerism).

- **3. TRUE coelom (schizocoel):** a mesoderm-lined body cavity filled with coelomic fluid that acts as a **HYDROSTATIC SKELETON** (lets them burrow without bones).
- **4. Triploblastic + bilateral + PROTOTOME:** three germ layers, symmetric body, development from the blastopore as mouth.
- **5. Circulation CLOSED + aortic arches:** a complete blood system; earthworm "hearts" = the aortic arches on segments 7-11; blood is red (chlorocruorin).
- **6. Excretion by NEPHRIDIA:** one pair per segment (metanephridia); **NERVOUS** system = a ventral nerve cord + a supra/sub-oesophageal "brain" ganglion.
- **7. The 3 classes: POLYCHAETA** (marine, **PARAPODIA** + many setae, separate sexes + trochophore larva; e.g. *Nereis*); **OLIGOCHAETA** (soil/fresh water, few setae, no parapodia, hermaphrodite; e.g. *Pheretima* / earthworm); **HIRUDINEA** (leeches, few/no setae, 2 **SUCKERS**, 21 segments; e.g. *Hirudo*).
- **8. Reproduction:** polychaetes have separate sexes + a larva; earthworms & leeches are **HERMAPHRODITES** (but self-fertilization is impossible - **CROSS-fertilization** + a cocoon from the clitellum).

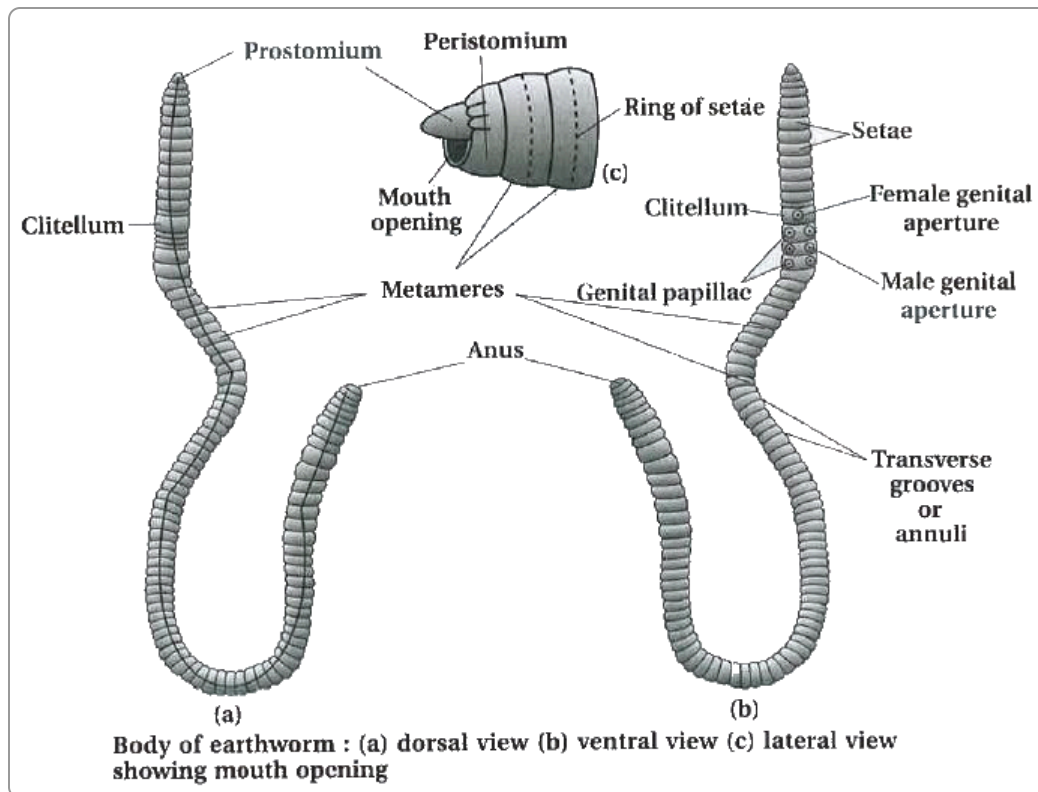


Diagram: earthworm external structure - label: segments (metameres), clitellum, setae, prostomium, mouth, anus

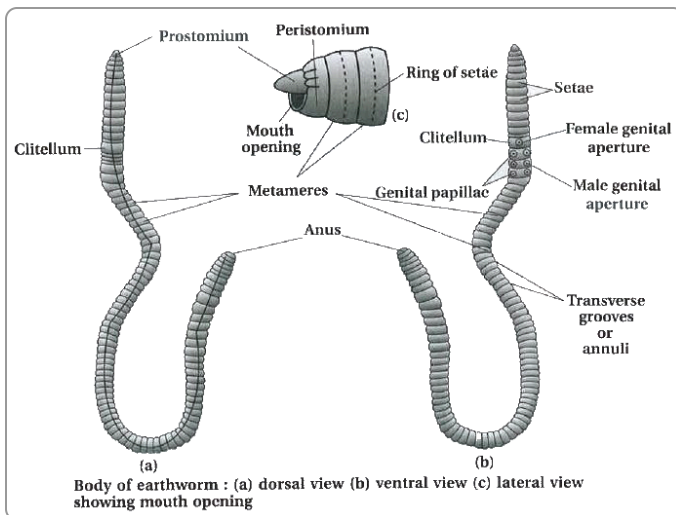
Examiner tip: "metamerism + true coelom + triploblastic + protostome + closed circulation + nephridia" = the 6 characteristic lines. Class ID: parapodia = polychaete; suckers = leech; setae only = oligochaete.

Q15 Describe the external structure of the earthworm (*Pheretima*) with a well-labelled diagram and mention 4 identifying facts. **8 marks**

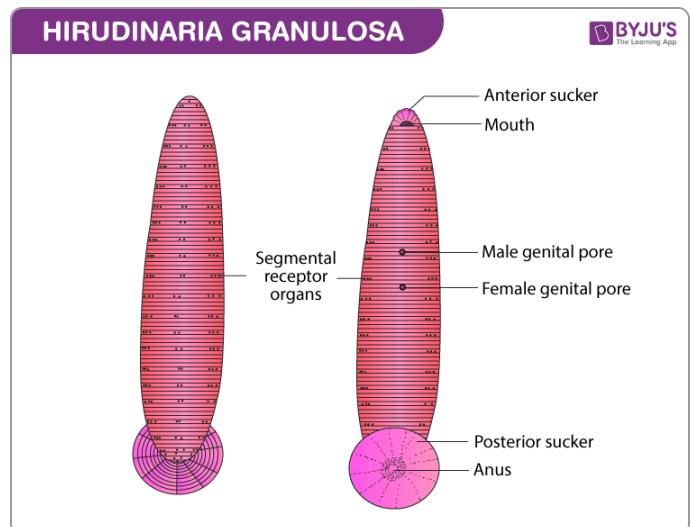
Earn the 8 marks: 1 labelled earthworm diagram (3) + 5 body-part descriptions (3) + 4 identifying "number" facts (2).

ANSWER

- **1. General form:** a long cylindrical worm of ~100-150 SEGMENTS; slightly curved, bluntly rounded anterior end.
- **2. Prostomium:** a small lobed appendage projecting over the mouth (before segment 1) - a sensory "snout".
- **3. Segments / metameres:** each ring bears 4 bundles of SETAE (bristles) that anchor the worm while it burrows.
- **4. Clitellum:** a prominent saddle-like band on SEGMENTS 34-37 (a wider "waist"); it secretes the COCOON that holds the eggs + sperm - marks the genital band.
- **5. Body cavity & setae:** a closed coelom + coelomic fluid = hydrostatic skeleton; setae push against soil = peristaltic movement.
- **6. Mouth & pharynx:** at the anterior, a mouth opens into a sucking pharynx (to pull in soil); the posterior end has the anus.
- **7. Genital pores:** two male + two female pores (the genital band ~ segments 14-27).
- **8. 4 identifying "number" facts:** (a) CLITELLUM = segments 34-37; (b) AORTIC ARCHES ("hearts") = segments 7-11; (c) ~100-150 segments; (d) hermaphrodite BUT cross-fertilization only (no self-fertilization).



Earthworm (Pheretima) external structure - label all 8 parts above



Compare with the leech (Hirudinea) - 21 segments, 2 suckers, no parapodia

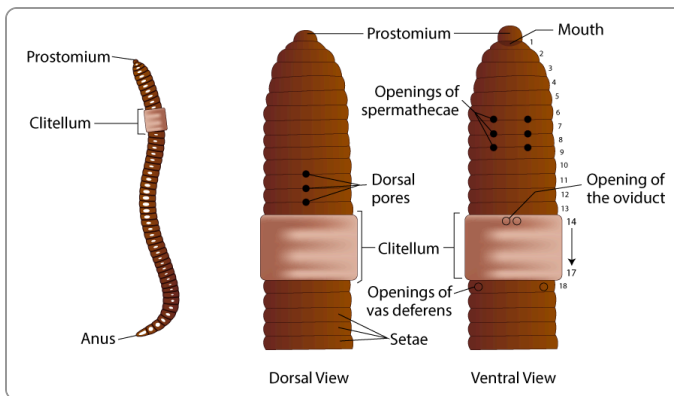
Examiner tip: the two "number facts" - clitellum 34-37 and aortic arches 7-11 - are the ones they test. Also: setae (bristles) = anchor; no parapodia = not a polychaete.

Q16 Explain metamerism and describe the internal organization of the earthworm (digestive, circulatory, excretory & nervous systems). **8 marks**

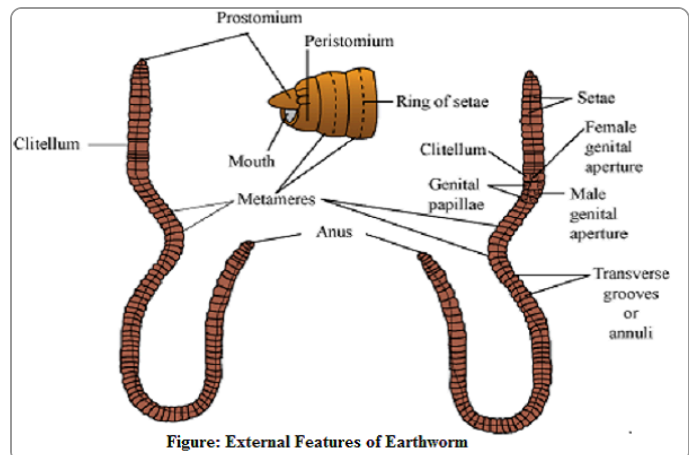
Earn the 8 marks: 1 internal-organ diagram (3) + 4 systems, each with 2 lines (4) + 1 metamerism + significance line (1).

ANSWER

- **1. Metamerism (define first):** segmentation where EXTERNAL body rings AND INTERNAL organs repeat; earthworms show HOMOMERIC (similar) metamerism; a "double metamer" = a somatic + a genital segment pair.
- **2. Digestive system:** mouth -> pharynx (sucking) -> oesophagus -> crop & GIZZARD (grinds soil; the calcium-cells live here) -> intestine (with a dorsal "intestinal caecum" for absorption) -> coelomic sac -> rectum -> anus. Earthworms eat soil & organic matter (humus).
- **3. Circulatory system (CLOSED):** a dorsal + ventral blood vessel + SETAL vessels; blood is pushed by the aortic arches ("hearts", segments 7-11); no white blood cells (phagocytes are coelomic).
- **4. Excretory system (NEPHRIDIA):** one pair of METANEPHRIDIA per segment (~100+ total); each = a flame-cup (nephrostome) -> coiled tubule -> a nephridiopore, filtering the coelomic fluid.
- **5. Nervous system:** a supra- & sub-oesophageal GANGLION ("brain") joined by a circum-oesophageal bridge; one VENTRAL NERVE CORD runs the length; each segment has a ganglion + perivisceral nerves; giant sensory & motor fibres for fast escape.
- **6. Reproductive system:** HERMAPHRODITE - two testis-cysts + two pair of ovaries; clitellum (34-37) builds the COCOON; CROSS-fertilization with an exchange-of-sperm (spermatheca stores the partner's sperm) - no self-fertilization.
- **7. Respiration & sense:** no gills/lungs - gas exchange across the MOIST body surface (why dry soil kills them); sense = light + touch (photoreceptors in the skin, no eyes).
- **8. Significance:** earthworms aerate the soil & add castings (plant food) - they are "the farm-yard fertilizers" of the soil.



Earthworm body plan - draw & label: coelom, aortic arches, gut, nephridia, nerve cord



Segmental organ arrangement (the metamerism detail)

Examiner tip: "metamerism" must be defined BEFORE the systems. The internal mark-set = coelom + closed circulation (aortic arches) + metanephridia + ventral nerve cord + hermaphrodite cross-fertilization.